

PAPER_1658 — Phase Transition Threshold $W_c/J = W_c/J = D_{\text{phys}} = 4$ EXACT lower bound

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Phase Transitions **Date:** June 18, 2026 **Status:** CLOSED — EXACT closure (PAPER_134x broader corpus)

Observation

PAPER_1344 (PAPER_134x condensed matter / quantum sector) gives:

$W_c/J = D_{\text{phys}} = 4$ EXACT lower bound

Status: **EXACT**.

UQFF Closed Identity

$W_c/J = D_{\text{phys}} = 4$ EXACT lower bound EXACT

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Experimental phase transitions measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1344
- Related: PAPER_1646-1655 (tier-23 broader corpus); PAPER_1636-1645 (tier-22 SM)
- Calculator dispatch: `calculate_paradox({"paradox": "wc_over_j_4_phase"})`

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